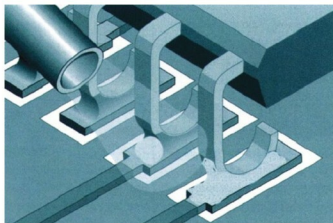


De-soldering Equipment

De-soldering is as much of a skill as soldering and is acquired over time. With the size of electronic components shrinking, the old-school methods used for de-soldering are not effective most of the time. But users generally understand the value of the right tools for de-soldering only after a number of bad experiences. The right tools can, however, make their rework effective and quick, or they could end up destroying their work beyond repair.



ponents from the PCB board. But these methods are really tedious and unproductive. The results only depend on your experience, and there is a high chance that you may damage the board beyond repair.

Over the years, these methods have

served their purpose very well. They have been in use for a long time and are very cost-effective, especially the use of solder braid, which is a ribbon of extremely fine copper strands woven together. It is usually supplied in small reels and comes in a variety of widths. The braid is pressed against a molten soldered joint, and capillary action draws the solder out of the joint and into the braid. The portion of the braid that contains solder is then cut off the reel and discarded. The solder braid is usually treated with a flux, so that the solder is drawn up more

effectively. But as the technology is changing and electronic components are shrinking in size, these methods have become less effective.

The prices of professional de-soldering tools have also come down over the past few years, which gives us every reason to buy one and save ourselves from all those losses that may occur with the tedious methods mentioned above.

Types of de-soldering stations

There is a wide variety of de-soldering tools available in the market, and each type is designed to cater to specific needs. Manufacturers offer different models in each category with various features and specifications, so selecting what's best for you can get confusing. You can address this issue in two stages—first decide on the type of de-soldering station you need and then pick a model from the selected category.

De-soldering stations can be divided into different types:

1. The first type is for de-soldering through-hole components and is called suction type de-soldering station.
2. For SMD (surface mount devices) components, you need a different de-soldering tool known as the hot-air rework system.
3. Another tool called the heated tweezer has a more targeted approach, just as its name suggests.
4. The above tools cannot de-solder BGA (ball grid array) components. BGA packages need BGA de-soldering station specifically designed for the purpose.

ANKIT GUPTA

De-soldering is required when any electronic component needs to be removed from a PCB, usually because it is faulty. Ideally, the aim is to remove all the solder from a component and detach the component from the PCB without damaging the de-soldered component, the neighbouring components or the PCB itself, either due to excessive heat or rough handling. There are also improperly-made soldered joints that are called dry joints, which are electrically noisy, unreliable and likely to get worse, in time. They may work well initially but can cause equipment failure at any time in the future. There will then come a time when you need to remove the solder from such a joint.

You can get a number of de-soldering guides on the Internet teaching old-school methods such as using a solder braid, a manual de-soldering pump, or the impact method to remove com-

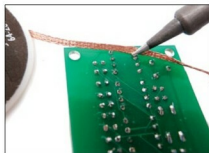


Fig. 1: De-soldering using a solder braid